

Domino: Hierarchical Benchmarking of LMMs for Reasoning Hallucinations in Multimodal Reasoning

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Abstract—This document describes the most common article elements and how to use the IEEEtran class with L^AT_EX to produce files that are suitable for submission to the IEEE. IEEEtran can produce conference, journal, and technical note (correspondence) papers with a suitable choice of class options.

Index Terms—Large multimodal models, hallucination, chain-of-thought reasoning, benchmark, multimodal reasoning, directed acyclic graph.

I. INTRODUCTION

RECENT advances in large language models (LLMs) and large multimodal models (LMMs) have significantly advanced artificial intelligence systems toward more general problem-solving capabilities. With the integration of visual perception, language understanding, and Chain-of-Thought (CoT) reasoning, current models are increasingly capable of handling challenging multimodal reasoning tasks. However, hallucination remains a widespread issue, where models may generate confident and plausible content that contradicts input evidence, established knowledge, or logical constraints. As CoT is widely used to enhance reasoning, hallucinations may also emerge within intermediate reasoning processes rather than only in final answers. Such reasoning-chain hallucinations undermine the reliability and trustworthiness of LMMs, making their comprehensive evaluation crucial for advancing reliable multimodal intelligence.

Current benchmarks for evaluating the reasoning abilities of LMMs still lack a systematic framework for assessing hallucinations in the reasoning process. Existing evaluation methods usually lack fine-grained diagnosis of different types of claims in the reasoning process, making it difficult to distinguish whether an error originates from visual perception, knowledge usage, or logical inference. Moreover, existing methods often regard reasoning chains as linear step sequences or flattened sets of claims, ignoring the fact that real reasoning processes are inherently non-linear and involve complex logical dependencies among different claims. Therefore, relying only on step-by-step checking or simple claim-level accuracy is insufficient to characterize critical errors in model reasoning and their influence on subsequent reasoning steps.

In addition, as shown in Table I, many reasoning-oriented benchmarks mainly rely on answer matching accuracy, where model performance is measured by whether the final output is consistent with the reference answer. Such an evaluation compresses a complex reasoning process into a binary judgment of

“correct” or “incorrect”, making it difficult to reveal whether the answer is supported by a reliable reasoning process. This issue is particularly important in complex multimodal reasoning tasks, where models may generate intermediate reasoning content that appears plausible but contains hidden errors. As a result, final-answer correctness alone cannot sufficiently reflect the quality of the reasoning process.

Meanwhile, a large portion of multimodal reasoning benchmarks still adopt multiple-choice question-answering formats. Although multiple-choice formats simplify automatic evaluation, candidate options may introduce information leakage, allowing models to exploit semantic cues, formatting patterns, or elimination strategies provided by the options. This weakens the ability of benchmarks to faithfully reflect the actual reasoning capabilities of models. Although some recent CoT-oriented benchmarks have begun to examine intermediate reasoning steps, they still struggle to systematically evaluate the importance of different reasoning errors and their influence on subsequent reasoning. Therefore, there remains a clear gap in accurately assessing the reliability of reasoning processes in real-world multimodal reasoning scenarios.

To bridge this gap, several key challenges need to be addressed: 1) relying solely on answer-matching accuracy fails to reveal errors in the reasoning process; 2) multiple-choice formats introduce information leakage and shortcut cues; 3) existing evaluations lack fine-grained diagnosis of hallucination types during reasoning; and 4) systematic metrics are still missing for evaluating how hallucinations accumulate and propagate throughout the reasoning process.

To address the above challenges, we introduce Domino-Bench, a new benchmark for evaluating hallucinations in open-ended multimodal reasoning processes. First, Domino-Bench reformulates multiple-choice questions into open-ended question-answering tasks, reducing option-induced shortcut cues and better reflecting real-world problem-solving scenarios where predefined candidate answers are unavailable. Second, the dataset covers six disciplines, enabling the evaluation of hallucinations in visual perception, disciplinary knowledge reasoning, and multi-step reasoning reliability.

Notably, Domino-Bench not only covers single-solution problems, but also explicitly considers multi-solution cases, with approximately 65.6% of the questions having more than one reasonable solution path. Rather than being limited to final-answer-level hallucination evaluation, Domino-Bench constructs verifiable intermediate reasoning processes and provides fine-grained annotations for process-level evaluation. On average, each question contains 22.1 verifiable claims, and each claim is annotated with its type, including perception,

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TABLE I
COMPARISON WITH EXISTING EVALUATION BENCHMARKS FOR LMMs

Benchmark	Disc.	Scale	Modality	Format	Multi-path	Chain Qual.	Claim Verif.	Hallu. Type	Dep. Model.	Err. Prop.
GSM8K	Single	8,792	Text	Open-ended	✗	✗	✗	✗	✗	✗
MATH	Single	12,500	Text	Open-ended	✗	✗	✗	✗	✗	✗
ARC-Challenge	Single	2,590	Text	MCQ	✗	✗	✗	✗	✗	✗
LogiQA	Single	8,678	Text	MCQ	✗	✗	✗	✗	✗	✗
MMLU	Multi	15,908	Text	MCQ	✗	✗	✗	✗	✗	✗
MMLU-Pro	Multi	12,032	Text	MCQ	✗	✗	✗	✗	✗	✗
GPQA	Multi	448	Text	MCQ	✗	✗	✗	✗	✗	✗
MMMU	Multi	11,500	Image-text	Mixed	✗	✗	✗	✗	✗	✗
MMBench	Multi	2,974	Image-text	MCQ	✗	✗	✗	✗	✗	✗
ScienceQA	Multi	21,208	Image-text	MCQ	✗	✗	✗	✗	✗	✗
M3CoTBench	Single	1,079	Image-text	QA/CoT	✗	✓	✗	✗	✗	✗
M ³ CoT	Multi	2,359	Image-Text	MC+CoT	✗	✓	✗	✗	✗	✗
VCR-Bench	Multi	1,034	Video-text	QA/CoT	✗	✓	△	P/R	✗	✗
MME-CoT	Multi	1,130	Image-text	QA/CoT	✓	✓	△	✗	✗	✗
MIRAGE	Multi	1,329	Image-text	QA/CoT	✗	✓	✓	R	✗	✗
MMReason	Multi	1,384	Image-text	Open-ended QA/CoT	✗	✗	✓	R	✗	✗
MM-THEBench	Multi	1,340	Image/Video-text	MCQ + Open-ended	✗	✓	✗	P/K/R	✗	✗
Domino-Bench	Multi	6,018	Image-text	Open-ended QA	✓	✓	✓	P/K/R	✓	✓

Note. **Disc.** indicates whether a benchmark covers a single discipline or multiple disciplines. **Multi-path** denotes whether the benchmark explicitly considers multiple valid solutions or reasoning paths for the same question. **Chain Qual.** denotes whether the benchmark evaluates the overall quality of the reasoning chain. **Claim Verif.** denotes fine-grained claim-level verification. **Hallu. Type** denotes hallucination types: perception (P), knowledge (K), and reasoning (R). **Dep. Model.** and **Err. Prop.** indicate reasoning dependency modeling and error propagation analysis, respectively.

knowledge, and reasoning. These detailed annotations provide reliable support for judging the correctness of model reasoning chains and analyzing different types of errors.

Different from existing works that usually treat CoT as a linear sequence of steps or a flattened set of claims, Domino-Bench further converts model-generated CoTs into directed acyclic graph (DAG) to explicitly model the logical dependencies among intermediate claims. In this graph structure, each node represents a verifiable intermediate claim, and each edge represents a logical dependency between claims. Through this representation, the reasoning process is no longer viewed as simple linear text, but is modeled as a structured derivation process composed of multiple interdependent intermediate conclusions. Therefore, reasoning errors are no longer treated as isolated local mistakes, but can be analyzed more carefully according to their positions, dependencies, and downstream influence in the reasoning structure. Based on this structure, we further design a DAG-based metric, whose computation process is illustrated in Fig. 1. The metric first judges whether each node is valid under its local semantics, then combines this local validity with the support inherited from its upstream nodes to compute the overall reliability of each node. Finally, the reliability of the conclusion node is used as the structured score of the entire reasoning chain. Compared with a flattened claim hit rate that treats all claims equally, the DAG-based metric leverages the graph structure to distinguish the roles of different claims in the reasoning process. Some key intermediate conclusions may support multiple downstream claims or serve as summary steps in later reasoning stages, and their correctness has a more significant impact on the final answer. Therefore, this structured evaluation more accurately reflects the contribution of key intermediate conclusions to the final

answer, providing a more faithful measure of the reasoning quality and reliability of reasoning-oriented VLMs.

Fine-grained process-level diagnosis not only helps reveal how hallucinations emerge and propagate, but also creates opportunities for selective intervention in the reasoning process. Motivated by this observation, we further introduce Counterpoise, a lightweight, training-free mitigation baseline. Unlike methods that obtain alternative reasoning trajectories through repeated sampling or complete regeneration, Counterpoise leverages conditional entropy to identify potentially unstable spans in the original reasoning trajectory and employs a View Router to selectively verify the visual evidence on which these spans rely. Only when the newly acquired evidence explicitly contradicts the original reasoning does Counterpoise preserve the unaffected reasoning prefix and locally repair the affected downstream process. This framework aims to provide a selective intervention strategy that balances reasoning performance against additional computational overhead, while demonstrating the potential of Domino-Bench to support research extending from hallucination diagnosis to mitigation.

Contributions:

(1) We propose **Domino-Bench**, a process-level benchmark for evaluating hallucinations in open-ended multimodal reasoning. It covers six theory-intensive disciplines and comprises 4,000 questions with 66,302 verifiable key claims. By accommodating multiple valid solution paths, Domino-Bench enables fine-grained diagnosis of hallucinations hidden in intermediate reasoning processes. (2) We develop a directed acyclic graph (DAG)-based evaluation framework and introduce **DAG-HS**, a dependency-aware hierarchical support metric that jointly models claim correctness and reasoning dependencies to measure the overall reliability of reasoning chains. Building

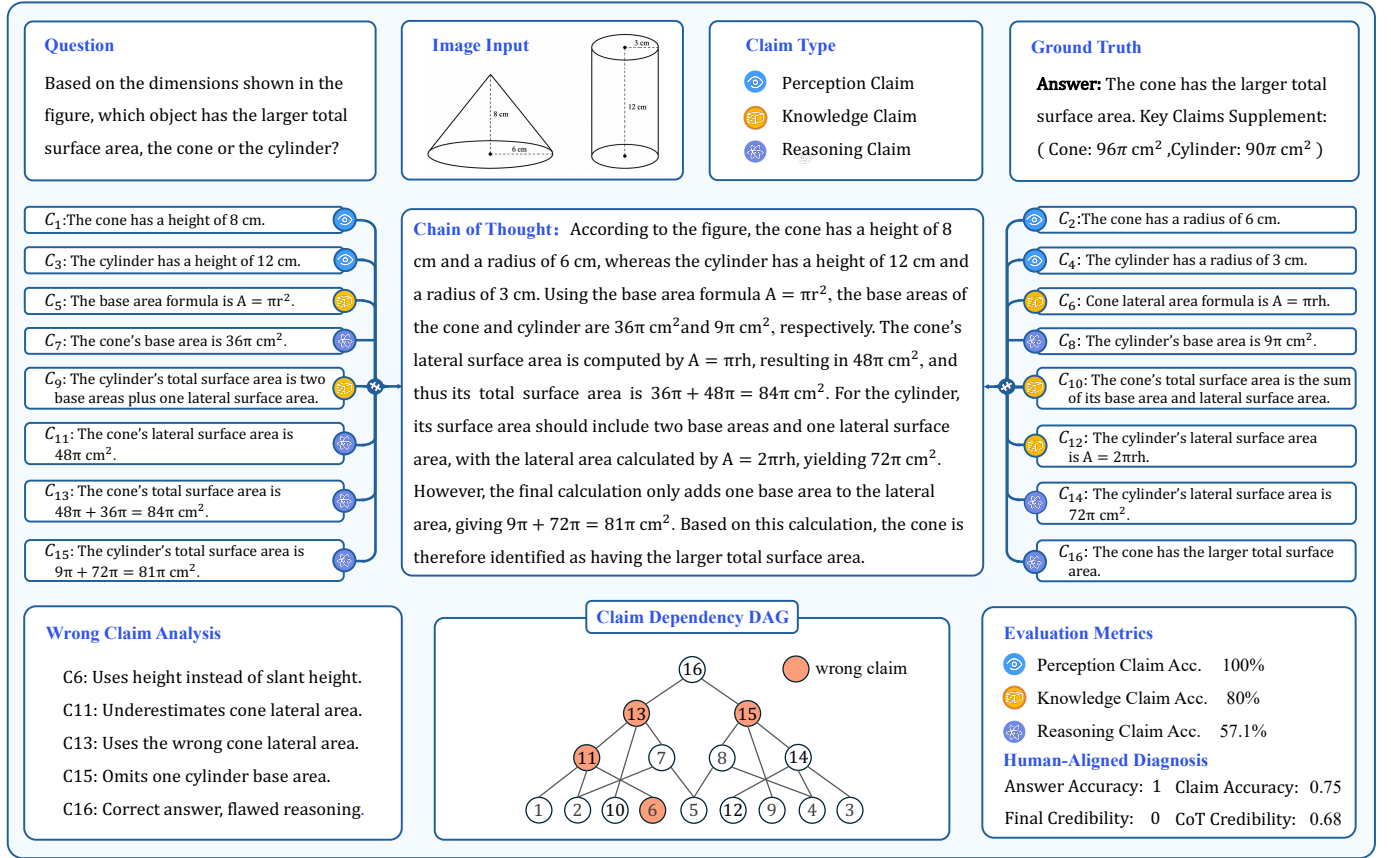


Fig. 1. Overview of our DAG-based recursive credibility metric. Given a multimodal input and a model-generated CoT response, we first decompose the reasoning process into claim nodes and annotate each claim as a perception, knowledge, or reasoning claim. We then construct a claim dependency DAG to capture the logical dependencies among claims. Our metric evaluates both local claim validity and recursive support credibility, enabling comparison between claim accuracy and CoT credibility, as well as between final answer correctness and final credibility.

on this framework, we further conduct a systematic analysis of process-level hallucinations, including the identification of critical reasoning errors and the characterization of hallucination propagation along reasoning dependencies. (3) We propose **Counterpoise**, a lightweight, training-free framework for selective reasoning repair. It leverages conditional entropy to locate potentially unstable spans in the original reasoning trajectory and employs a View Router to selectively verify the relevant visual evidence. Only when an explicit contradiction is identified does Counterpoise reuse the unaffected reasoning prefix and locally repair the downstream reasoning process. (4) We extensively evaluate 19 representative LMMs on Domino-Bench, including frontier commercial models such as GPT-5.6-sol, Gemini-3.1-Pro, and Grok-4.5, as well as advanced open-weight models from the Gemma 4, GLM-4.1V, and Qwen3-VL families, covering diverse parameter scales and multimodal architectures.

II. RELATED WORK

A. Large Model Reasoning

Thanks to the rapid development of large language models (LLMs) and multimodal large language models (MLLMs), recent models have demonstrated strong capabilities in knowledge modeling, cross-domain multitasking, and complex reasoning. Reasoning-oriented models represented by OpenAI-o1

further show that models can approach expert-level problem analysis and multi-step deduction abilities in challenging tasks such as mathematics and programming [1]–[4]. Inspired by these advances, the research focus of MLLMs has gradually moved beyond basic image-text understanding and visual question answering toward more complex multimodal reasoning tasks [6], [7], [10], [11]. To enhance multimodal reasoning capabilities, existing studies have explored several directions, including prompt-based reasoning, process supervision and reward modeling, search- and planning-based reasoning, and learning-based reasoning. Prompt-based methods typically introduce Chain-of-Thought (CoT) to encourage models to explicitly generate intermediate reasoning steps before producing final answers, thereby improving their performance on complex visual question answering and multimodal reasoning tasks [5]–[7]. Building upon this line of work, process supervision and process reward models (PRMs) further focus on the quality of intermediate reasoning steps, providing finer-grained feedback beyond final-answer correctness [8], [9]. Such process-level feedback provides important support for search-based planning and reinforcement learning, enabling models to compare, select, and optimize among multiple candidate reasoning paths. Search- and planning-based methods further alleviate the instability of single-path generation by exploring multiple candidate reasoning chains. For example,

Mulberry incorporates collective knowledge into Monte Carlo Tree Search (MCTS) to search and select more reliable reasoning paths, thereby enhancing the reasoning and reflection abilities of MLLMs [10]. In addition, inspired by DeepSeek-R1, recent works have begun to apply reinforcement learning to the multimodal domain, using verifiable rewards, process feedback, or task-level rewards to further improve the reasoning ability of MLLMs in open-ended visual problems [4], [11], [12]. Based on these advances, we introduce Domino-Bench to quantify the cross-domain adaptation, perceptual understanding, and reliable reasoning capabilities of VLMs in multimodal and multi-domain tasks.

B. LMMs Benchmark

Evaluation benchmarks play a pivotal role in advancing the development of large multimodal models (LMMs). Early reasoning benchmarks, such as GSM8K, MATH, ARC-Challenge, LogiQA, MMLU, MMLU-Pro, and GPQA, have systematically evaluated language models from the perspectives of mathematical reasoning, commonsense reasoning, logical inference, and multi-disciplinary knowledge understanding [13]–[19]. These benchmarks provide important foundations for measuring text-based reasoning abilities. However, they are primarily designed for text-only tasks, lacking visual inputs and cross-modal information, and thus are insufficient for evaluating the perceptual understanding and cross-modal reasoning capabilities required by LMMs in real multimodal scenarios. To evaluate multimodal understanding, subsequent benchmarks such as MMMU, MMBench, and ScienceQA introduce image-text tasks covering visual perception, scientific question answering, multi-disciplinary knowledge understanding, and general multimodal problem solving [20]–[22]. These benchmarks have significantly promoted the development of LMMs and provided unified testbeds for comparing different models. However, as shown in Table I, many existing multimodal benchmarks still heavily rely on multiple-choice or true/false formats. Although such formats facilitate automatic evaluation, they may introduce option-induced shortcuts, allowing models to obtain correct answers through elimination, guessing, or memorization rather than genuine reasoning. Recently, as LMMs have evolved from basic image-text understanding toward complex multi-step reasoning, several benchmarks have begun to focus on multimodal Chain-of-Thought reasoning processes. M3CoTBench, VCR-Bench, MME-CoT, MIRAGE, MMReason, and MM-THEBench evaluate intermediate reasoning processes from different perspectives, including reasoning-chain quality, intermediate-step correctness, and hallucination types [23], [24], [26]–[29]. These works mark an important shift from final-answer evaluation toward process-level evaluation, enabling more fine-grained analysis of error patterns in multimodal reasoning chains. In comparison, Domino-Bench adopts a more comprehensive evaluation protocol: it covers six theory-intensive disciplines, reformulates multiple-choice problems into open-ended question-answering tasks, and uses fine-grained, verifiable claims to evaluate model reasoning processes. The benchmark emphasizes cross-domain adaptation, fine-grained

perceptual understanding, and reliable multi-step reasoning, aiming to provide a more detailed and in-depth evaluation for reasoning-oriented VLMs.

C. LMMs Hallucination in Multimodal Reasoning

Hallucinations generated by large models have attracted widespread attention. In the NLP community, hallucinations generally refer to outputs that are inconsistent with facts, input contexts, or internal logic, and are commonly discussed from factual and faithful perspectives [30]. Compared with text-only LLMs, hallucinations in vision-language models (VLMs) are more complex, since responses must be grounded not only in language contexts but also in visual inputs. In multimodal reasoning scenarios, models further need to align visual signals, textual semantics, domain knowledge, and logical derivations. Therefore, hallucination evaluation for reasoning-oriented VLMs should consider not only cross-modal grounding, but also how errors emerge and propagate throughout the reasoning process.

Existing studies mainly evaluate VLM hallucinations from perception-oriented perspectives. For example, POPE [31] evaluates object hallucination by testing whether models incorrectly affirm nonexistent objects in images; Hallusion-Bench [32] analyzes the entanglement between language hallucination and visual illusion; AMBER [33] evaluates multimodal hallucinations from existence, attribute, and relation dimensions; and MMHal-Bench [34] examines whether model responses are faithful to visual evidence and factual content. These benchmarks have advanced hallucination evaluation for VLMs, but they primarily focus on perception-level cross-modal inconsistencies and pay limited attention to intermediate errors and their dependency relations in complex reasoning chains. Recent studies further show that hallucinations are not always isolated mistakes, but can be amplified by the generation context and propagate along premise–conclusion dependencies in reasoning chains [35]–[38].

As VLMs are increasingly applied to complex multi-step reasoning tasks, benchmarks such as MIRAGE [27] and MM-THEBench [29] have begun to investigate hallucinations in multimodal reasoning chains from step-level, claim-level, or hallucination-type perspectives. However, existing studies still mainly focus on hallucination categories, local reasoning errors, or textual CoT diagnosis, while lacking systematic modeling of the logical dependencies among intermediate claims. This makes it difficult to analyze how perception, knowledge, and reasoning errors propagate along multimodal reasoning chains. In comparison, Domino-Bench converts model-generated CoTs into directed acyclic graphs to explicitly model dependencies among claims, enabling a more faithful assessment of reasoning quality, reasoning reliability, and hallucination propagation in reasoning-oriented VLMs.

III. DOMINO-BENCH

A. Benchmark Construction

In this section, we introduce the data construction pipeline developed for Domino-Bench. The pipeline aims to construct an open-ended multimodal multi-step reasoning benchmark

that requires models to jointly leverage visual perception, disciplinary knowledge, and logical reasoning, rather than relying on candidate options, shallow visual recognition, or simple answer matching. As summarized in Fig. 2, the construction process consists of candidate dataset collection, target sample filtering, open-ended question reformulation, CoT generation with reflection-based refinement, claim decomposition and KPR annotation, and final data curation and bilingualization.

1) *Candidate Dataset Collection*: We first collect candidate samples from multiple public multimodal disciplinary reasoning datasets, including eee-bench, GeoSQA, scemqa, mm-math, physreason, sciverse, seephy, mathvision, multi-physics, CMM-math, Geometry3k, EMMA, Phyx, geoqa-plus, and AI2D. These datasets cover six theory-intensive disciplines, including mathematics, physics, electrical engineering, chemistry, biology, and geography. They also contain diverse visual inputs, such as natural images, geometric diagrams, scientific illustrations, maps, tables, circuit diagrams, function plots, and structured symbolic figures. These data sources provide a broad foundation for constructing multimodal reasoning tasks with both disciplinary diversity and visual complexity.

2) *Target Sample Filtering*: After obtaining the initial candidate pool, we further filter samples that are suitable for Domino-Bench. Specifically, we use GPT-5.4 to automatically assess each candidate question according to predefined filtering criteria. First, the question must require information from the corresponding image and cannot be fully solved from the textual prompt alone. Second, the question should involve a multi-step reasoning process rather than pure perception, simple visual reading, direct value lookup, or single-step matching. Through this filtering process, we remove samples that mainly test basic visual recognition or shallow information extraction and retain problems that require the integration of visual evidence, disciplinary knowledge, and multi-step derivation.

3) *Open-ended Question Reformulation*: Many existing multimodal disciplinary reasoning datasets adopt multiple-choice or true/false formats. Directly using these questions may introduce option-induced shortcuts, allowing models to obtain correct answers through elimination, guessing, or semantic matching without performing the intended reasoning process. To reduce the influence of candidate options, we reformulate non-open-ended questions into open-ended question-answering tasks.

Specifically, we rewrite each question according to the semantic intent of the original query and the structure of its answer options. When the options belong to the same semantic category, such as numerical values, object categories, or geographical locations, we reformulate the question to directly request the target result. When the options are heterogeneous or correspond to different reasoning objectives, we reorganize the question according to its central inquiry while removing explicit candidate answers and preserving the original problem-solving objective. Through this process, Domino-Bench better reflects open-ended problem solving, requiring models to actively generate both answers and reasoning processes rather than relying on predefined options.

4) *CoT Generation and Reflection-based Refinement*: For

each open-ended question, we construct one or more high-quality reference reasoning processes. Specifically, we provide the question text and corresponding image to GPT-5.4 and prompt it to generate multiple possible solution paths when applicable. Each candidate solution path is subsequently expanded with the necessary intermediate reasoning details to form a complete, coherent, and verifiable chain-of-thought (CoT).

To improve the reliability of the reference reasoning process, we adopt a reflection-based refinement mechanism. GPT-5.4 is instructed to inspect each generated CoT from three perspectives: visual perception, knowledge content, and logical reasoning. The inspection covers potential visual misreadings, incorrect knowledge usage, improper application of formulas or concepts, invalid derivation steps, and inconsistencies between the reasoning process and final answer.

If the answer derived from the CoT is consistent with the ground-truth answer, the reasoning chain is retained. Otherwise, the identified problems are provided to the model as feedback for revision and supplementation. This refinement process is repeated for at most three rounds. Samples whose reasoning processes remain inconsistent with the ground-truth answer are removed from the benchmark.

5) *Claim Decomposition and KPR Annotation*: After obtaining reliable reference CoTs, we further decompose each reasoning process into fine-grained and verifiable claims. Each claim represents a minimal visual observation, knowledge statement, semantic judgment, intermediate conclusion, or final conclusion in the reasoning process. These claims serve as the basic units for subsequent DAG construction and process-level evaluation.

Following the taxonomy defined in Sec. IV-A, each claim is assigned a KPR label: perception, knowledge, or reasoning. Claims involving image reading, visual localization, relation recognition, or visual grounding are labeled as perception. Foundational claims involving textual conditions, disciplinary concepts, formulas, theorems, rules, or factual knowledge are labeled as knowledge. Claims representing intermediate conclusions, key deductions, or final conclusions derived from existing claims are labeled as reasoning. This annotation enables fine-grained diagnosis of model capabilities and errors at the perception, knowledge, and reasoning levels.

6) *Final Curation and Bilingualization*: After claim decomposition and annotation, we perform a final round of quality control and distribution calibration. We remove duplicated samples, ambiguous questions, inconsistent annotations, and instances whose reference reasoning processes cannot be reliably verified.

Rather than enforcing a strictly uniform number of samples across disciplines, we prioritize the quality and reasoning complexity of each retained instance. The six disciplines are used to provide broad cross-domain reasoning coverage, while the final curation process mainly controls key evaluation dimensions such as problem difficulty and reasoning-path characteristics. In particular, we preserve sufficient numbers of both single-path and multi-path problems to support evaluation under different feasible reasoning strategies.

Finally, each retained sample is constructed in both Chinese

TABLE II
OVERALL STATISTICS OF THE DOMINO BENCHMARK

Statistics	Number
Total Questions	6,000
Total Claim Annotations	121,100
Average Claims per Question	20.18 (≈ 21)
Subject Domains	6
Language Versions	2 (En., Zh.)
Questions with Multiple Solutions	828 (13.80%)
Questions with One Standard Solution	5,172 (86.20%)
Mathematics Questions	1,670 (27.83%)
Physics Questions	800 (13.33%)
Chemistry Questions	900 (15.00%)
Biology Questions	700 (11.67%)
Geography Questions	1,230 (20.50%)
Circuit Questions	700 (11.67%)
Easy Questions	1,860 (31.00%)
Medium Questions	3,240 (54.00%)
Hard Questions	900 (15.00%)

and English. During bilingualization, we preserve the semantic content, problem conditions, disciplinary terminology, and reasoning structure of the original sample, enabling Domino-Bench to support open-ended multimodal reasoning evaluation under both Chinese and English language settings.

B. Dataset Statistics

Domino-Bench contains 6,018 open-ended multimodal reasoning problems spanning six theory-intensive disciplines: mathematics, physics, electrical engineering, chemistry, biology, and geography. The disciplinary composition is not designed to be strictly uniform. Instead, each discipline provides a substantial collection of high-quality problems, and the complete benchmark is intended to evaluate the cross-domain generalization ability of multimodal reasoning models. Detailed statistics for the six disciplines are presented in Table ??.

The benchmark contains a total of 122,670 fine-grained annotated claims, corresponding to approximately 20.4 claims per problem on average. Each claim is assigned a perception, knowledge, or reasoning label and is further organized according to its dependency relationships in the reasoning DAG. The resulting annotations provide dense supervision and evaluation units for measuring both local claim correctness and the structural reliability of complete reasoning processes.

In addition to problems with a single primary solution path, Domino-Bench preserves problems that admit multiple reasonable reasoning paths. Multi-path problems account for approximately 15% of the complete benchmark. Rather than requiring models to reproduce a single reference CoT, these samples allow different valid reasoning strategies to be evaluated according to their claim validity and dependency consistency.

We also control the distribution of problem difficulty during final data curation to prevent the benchmark from being dominated by either trivial problems or excessively difficult instances. The resulting benchmark contains problems with

diverse levels of reasoning complexity, ranging from relatively direct multimodal derivations to problems involving long-range dependencies and multiple intermediate conclusions.

Each problem is provided in both Chinese and English, resulting in two semantically aligned language versions. The bilingual construction supports evaluation under different language settings and enables further analysis of whether multimodal reasoning performance remains consistent across languages.

IV. PROCESS-LEVEL HALLUCINATION EVALUATION AND MITIGATION

A. DAG Representation and Hallucination Taxonomy

Open-ended multimodal reasoning is usually not a simple linear sequence of steps, but a hierarchical derivation process gradually organized by multiple interdependent claims. To characterize such a logical hierarchy, Domino represents model-generated CoTs as directed acyclic graphs (DAGs). In each DAG, every node denotes a verifiable claim, and each edge denotes a logical dependency between claims. In general, bottom-layer leaf nodes correspond to textual information from the problem statement, visual perception information, and domain knowledge; intermediate nodes represent conclusions derived from existing claims; and the final node corresponds to the final answer obtained by integrating multiple intermediate conclusions. Through this representation, the reasoning process is no longer treated as flattened text, but is modeled as a hierarchical reasoning structure composed of perceptual evidence, knowledge support, intermediate derivations, and final conclusions.

Based on the above DAG representation, Domino defines hallucination types according to the content of each claim and its position in the reasoning graph. Bottom-layer claims related to reading, recognizing, or grounding visual inputs are regarded as perception claims; bottom-layer supporting claims related to disciplinary concepts, formulas, theorems, rules, factual knowledge, or textual conditions in the problem statement are regarded as knowledge claims; and intermediate conclusions, key conclusions, or final conclusions derived from existing parent claims are regarded as reasoning claims. When these claims are invalid, they correspond to perception hallucination, knowledge hallucination, and reasoning hallucination, respectively. This definition directly aligns the hallucination taxonomy with the logical hierarchy in the DAG, so that the type of each claim depends not only on its semantic content, but also on the role it plays in the reasoning graph.

1) *Perception Hallucination*: Perception hallucination refers to cases where a perception claim is inconsistent with the visual evidence, meaning that the model fails to accurately understand, read, or locate key information in the visual input. Such errors mainly arise from misrecognizing, misinterpreting, or fabricating visual content, reflecting failures in visual perception and visual grounding.

2) *Knowledge Hallucination*: Knowledge hallucination refers to cases where a knowledge claim is inconsistent with relevant disciplinary knowledge, formulas, rules, or problem conditions, meaning that the model fails to correctly invoke or

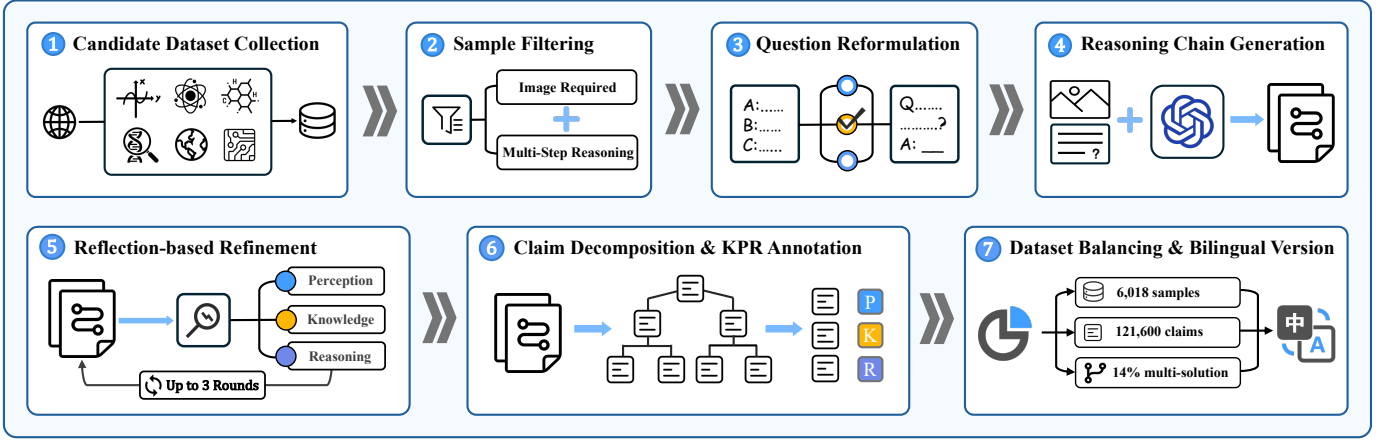


Fig. 2. Our benchmark construction pipeline. 1: Candidate samples are collected from multiple multimodal disciplinary reasoning datasets. 2: Target samples requiring visual information and multi-step reasoning are filtered. 3: Non-open-ended questions are reformulated into open-ended QA tasks. 4: Multiple candidate CoTs are generated using GPT-5.4-xhigh. 5: CoTs are refined through reflection-based feedback until the final answers align with the ground truth. 6: CoTs are decomposed into fine-grained claims with KPR annotations. 7: The final dataset is balanced and bilingual versions are constructed, resulting in 6,018 samples and 122,670 annotated claims.

apply the domain knowledge required to solve the problem. Domino-Bench covers six theory-intensive disciplines, including mathematics, physics, electrical engineering, chemistry, geography, and biology. Such errors are typically manifested as incorrectly citing or applying formulas, theorems, concepts, rules, or domain-specific facts, reflecting failures in understanding and using disciplinary knowledge.

3) *Reasoning Hallucination*: Reasoning hallucination refers to cases where a reasoning claim cannot be correctly derived from its parent claims, meaning that the model fails to obtain a correct and valid conclusion based on existing claims during multi-step reasoning. Here, existing claims include visual perception information, judgments supported by disciplinary knowledge, and intermediate conclusions produced by previous reasoning steps. Such errors mainly arise from failures in computation, derivation, or logical composition. Unlike knowledge hallucination, reasoning hallucination does not necessarily originate from incorrect formulas or domain knowledge; instead, it occurs when the model fails to reason correctly even though the existing information itself is valid.

By combining DAG representation with hallucination taxonomy, Domino-Bench describes the content types, logical hierarchy, and derivation relations within a reasoning chain under a unified structure. Bottom-layer nodes characterize whether the model reliably understands and uses the problem statement, visual evidence, and domain knowledge, while intermediate and final nodes characterize whether the model can perform valid derivations based on existing claims. This provides a foundation for subsequent claim-level annotation and hierarchy-aware evaluation.

B. Hierarchy-Aware Recursive Credibility Evaluation

To move beyond flat claim-level correctness statistics, Domino introduces a hierarchy-aware recursive credibility metric based on the DAG structure of model-generated reasoning chains. As illustrated in Fig. 1, given a multimodal input and a model-generated CoT response, we first decompose

the reasoning process into atomic claim nodes and annotate each claim as a perception, knowledge, or reasoning claim. We then construct a claim dependency DAG, where nodes represent verifiable claims and directed edges represent logical dependencies among claims.

Existing process-level benchmarks have recognized that final-answer correctness alone is insufficient for evaluating reasoning ability, since a correct answer may still be supported by an erroneous or unreliable intermediate process. However, we argue that this principle should also be applied recursively to the intermediate reasoning process itself: the final answer is a factual claim that should be supported by preceding steps, and each intermediate claim is likewise a factual or inferential claim that should be supported by its own premise claims. Therefore, evaluating intermediate claims should not only determine whether they are locally correct in isolation, but also examine whether they can be validly derived from reliable upstream premises. In other words, even if a claim is locally correct, its rationality and credibility in the current reasoning chain remain questionable if it is not grounded in reliable premises or cannot be correctly inferred from its premise claims. This reflects the recursive nature of multi-step reasoning: the credibility of a claim depends not only on its own local validity, but also on the credibility of the upstream reasoning subgraph that supports it. Based on this insight, we introduce recursive credibility evaluation.

Given a reasoning DAG $G = (V, E)$, where each node $v \in V$ denotes a verifiable claim and each directed edge denotes a dependency relation between claims, we define three quantities for each node: local validity $z(v)$, recursive support credibility $U(v)$, and propagated credibility $R(v)$.

Local validity. The local validity $z(v) \in \{1, 0\}$ indicates whether node v is locally valid. For perception and knowledge claims, $z(v)$ reflects whether the claim is supported by the visual input, the problem statement, or reliable disciplinary knowledge. For reasoning claims and the final answer node, $z(v)$ reflects whether the current claim can be correctly derived

from its premise claims, assuming that these premise claims hold.

Recursive support credibility. For a non-root node v , its recursive support credibility is defined as the weighted aggregation of the propagated credibility of its premise nodes:

$$U(v) = \sum_{u \in \text{Pre}(v)} w_{u,v} R(u),$$

where $\text{Pre}(v)$ denotes the set of premise nodes of v , and $w_{u,v}$ denotes the dependency weight from premise node u to node v . In our implementation, we adopt uniform weights:

$$w_{u,v} = \frac{1}{|\text{Pre}(v)|}.$$

For root nodes, since they do not have premise nodes, we set $U(v) = 1$. Intuitively, $U(v)$ measures how credible the upstream supporting subgraph of node v is.

Propagated credibility. The propagated credibility of node v is defined as:

$$R(v) = z(v) \cdot U(v).$$

Thus, $R(v)$ measures the credibility of a claim after considering both its own local validity and the recursive credibility of its supporting subgraph. Even if a claim is locally valid, it may still receive a lower propagated credibility score if it depends on unreliable upstream claims. This recursive formulation allows Domino to capture how errors in lower-level perception, knowledge, or reasoning claims affect downstream conclusions.

Based on $R(v)$, we define the CoT credibility score as:

$$\text{CoTCredibility} = \frac{1}{|V^*|} \sum_{v \in V^*} R(v),$$

where O denotes the final answer node, and $V^* = \text{Anc}(O) \cup \{O\}$ denotes the reasoning subgraph consisting of the final answer node and all its ancestor nodes. Since each DAG in Domino is constructed around the final answer, CoT Credibility measures the overall credibility of the reasoning process that supports the final answer.

We further define the final credibility score as:

$$\text{FinalCredibility} = R(O),$$

which evaluates whether the final answer is not only locally valid, but also supported by a credible upstream reasoning chain. Together, CoT Credibility and Final Credibility provide complementary evaluations of model reasoning ability from the perspectives of overall reasoning-chain credibility and final-conclusion credibility.

C. Counterpoise: A Training-Free Framework for Selective Reasoning Repair

Our process-level evaluation treats a reasoning failure as a localized event whose influence may propagate to its downstream conclusions. This perspective suggests that mitigation need not discard an otherwise reliable trajectory. Instead, an intervention should identify a potentially unstable segment, verify the evidence on which it depends, and revise only the affected continuation. Based on this principle, we introduce

Counterpoise, a lightweight, training-free framework for selective reasoning repair.

Unlike methods that sample multiple complete trajectories or regenerate the response from scratch, Counterpoise first obtains a single standard reasoning trajectory and preserves it by default. A revision is attempted only when newly acquired visual evidence explicitly contradicts a revisable fact in the original reasoning. Counterpoise requires neither parameter updates nor a separately trained auxiliary model. It also does not construct a reasoning DAG at inference time; rather, it operationalizes the same process-level principle through entropy-guided localization and contradiction-gated local replay.

1) *Initial Reasoning and Problem Setup:* Given an image I , a question q , and an LMM f_θ , Counterpoise first performs the same direct inference as the native baseline:

$$y^{(0)} = f_\theta(I, q; \mathcal{P}_{\text{direct}}), \quad (1)$$

where $y^{(0)} = (y_1^{(0)}, \dots, y_T^{(0)})$ is generated with greedy decoding. We parse from it an initial reasoning trajectory $R^{(0)}$ and final answer $a^{(0)}$:

$$(R^{(0)}, a^{(0)}) = \text{Parse}(y^{(0)}). \quad (2)$$

When the final answer is not directly parseable, a short text-only recovery prompt extracts the answer from the existing output without re-reading the image or re-solving the problem. If recovery still fails, Counterpoise returns the original output and terminates. This design ensures that every subsequent change is attributable to selective verification rather than to a different initial prompt or stochastic sampling.

2) *Entropy-Guided Localization of Unstable Segments:* After the initial generation, we perform one teacher-forced forward pass over $y^{(0)}$. At token position t , the model produces

$$p_t(v) = p_\theta(v \mid I, q, y_{<t}^{(0)}), \quad v \in \mathcal{V}, \quad (3)$$

where $\mathcal{V}$ denotes the vocabulary. We quantify local uncertainty using conditional entropy:

$$H_t = - \sum_{v \in \mathcal{V}} p_t(v) \log p_t(v). \quad (4)$$

A high value of H_t is treated only as a localization signal, rather than direct evidence of an error. Because individual subword tokens are often semantically incomplete, we map tokens back to their lexical spans and construct candidate windows containing at most three adjacent lexical tokens. For a valid window $s = (y_i^{(0)}, \dots, y_j^{(0)})$, its score is

$$S(s) = \frac{1}{j - i + 1} \sum_{t=i}^j H_t, \quad 1 \leq j - i + 1 \leq 3. \quad (5)$$

Windows crossing sentence boundaries, containing only stopwords or tokenizer fragments, or overlapping the final-answer marker are discarded. We rank the remaining windows by $S(s)$ and retain the top K non-overlapping candidates:

$$\mathcal{S}_K = \text{TopK}_{s \in \mathcal{S}_{\text{valid}}} S(s), \quad K = 5. \quad (6)$$

This step uses only the model's predictive distribution and deterministic lexical rules, avoiding external claim decomposition or learned error scoring.

3) *View Router and Answer-Blind Visual Verification*: The prompt-based View Router receives $(q, R^{(0)}, a^{(0)}, \mathcal{S}_K)$ and returns

$$(r, f, u) = \text{Route}\left(q, R^{(0)}, a^{(0)}, \mathcal{S}_K\right), \quad (7)$$

where f is a potentially fragile visual fact in the original trajectory, u specifies the object, region, or relation to inspect, and

$$r \in \{\text{BOTTOM_UP}, \text{TOP_DOWN}, \text{NORMAL}\} \quad (8)$$

is the selected inspection view.

BOTTOM_UP targets local attributes such as text, numbers, colors, symbols, counts, and arrow directions. TOP_DOWN targets global structures such as paths, topology, connectivity, and multi-object relations. NORMAL is used when uncertainty is distributed or does not clearly belong to either category. Only one view is executed, thereby directing the additional visual computation toward the suspected source of instability.

Before visual verification, Counterpoise protects facts that should not be overwritten by an image interpretation. We define

$$A(q, f) \in \{\text{PROTECTED}, \text{REVISABLE}\}. \quad (9)$$

A fact is PROTECTED if it is explicitly specified in the question, follows directly from a mathematical definition, or is an exact consequence of a theorem whose premises are fully stated in the text. Facts whose validity still depends on visually inferred labels, positions, or geometric relations remain REVISABLE. The corresponding authority gate is

$$g_{\text{auth}} = \mathbb{I}[A(q, f) = \text{REVISABLE}]. \quad (10)$$

Protected facts trigger an early exit, preventing uncertain visual interpretations from overriding authoritative textual premises.

For a revisable fact, the selected target is re-read under a neutral prompt:

$$E = f_{\theta}(I, q, u; \mathcal{P}_r, \mathcal{P}_{\text{neutral}}). \quad (11)$$

The re-reading prompt includes neither the value asserted by f nor the initial answer $a^{(0)}$. It asks only for directly visible evidence and requires an UNCERTAIN response when the target cannot be resolved. This answer-blind design reduces confirmation bias during verification.

4) *Contradiction-Gated Local Reasoning Repair*: The acquired evidence is compared with the fragile fact using

$$C(q, f, E) \in \{\text{SUPPORTS}, \text{CONTRADICTS}, \text{UNCERTAIN}\}. \quad (12)$$

CONTRADICTS is returned only when the image directly establishes an incompatible label, number, count, direction, position, connection, or relation. Additional but non-conflicting detail, evidence requiring substantial external inference, and ambiguous observations are assigned UNCERTAIN. Both SUPPORTS and UNCERTAIN preserve the original trajectory. The contradiction gate is therefore

$$g_{\text{con}} = \mathbb{I}[C(q, f, E) = \text{CONTRADICTS}]. \quad (13)$$

When an explicit contradiction is established, Counterpoise identifies the earliest reasoning step k that uses f . Writing the

original trajectory as $R^{(0)} = (r_1^{(0)}, \dots, r_m^{(0)})$, we retain the unaffected prefix and regenerate only the dependent suffix:

$$R_{<k}^{(1)} = R_{<k}^{(0)}, \quad (14)$$

$$R_{\geq k}^{(1)} = f_{\theta}\left(q, R_{<k}^{(0)}, f, E; \mathcal{P}_{\text{local}}\right), \quad (15)$$

$$R^{(1)} = R_{<k}^{(0)} \oplus R_{\geq k}^{(1)}, \quad (16)$$

where $\oplus$ denotes trajectory concatenation. The local replay does not re-read the image; it receives only the problem, the preserved prefix, the contradicted fact, and the newly grounded evidence. Consequently, unrelated visual facts and already reliable reasoning steps are less likely to be altered.

Finally, the candidate answer $a^{(1)}$ is accepted only if the revised output is parseable and passes deterministic answer-safety checks. These checks reject empty or meta-level outputs, anomalously verbose answers, and changes that violate the expected answer type, such as replacing a numerical answer with a non-numerical string. Let g_{parse} and g_{safe} denote the parsing and answer-safety gates, respectively. The overall acceptance decision is

$$g_{\text{accept}} = g_{\text{auth}} g_{\text{con}} g_{\text{parse}} g_{\text{safe}}, \quad (17)$$

and the final output is

$$(R^*, a^*) = \begin{cases} (R^{(1)}, a^{(1)}), & g_{\text{accept}} = 1, \\ (R^{(0)}, a^{(0)}), & \text{otherwise.} \end{cases} \quad (18)$$

Thus, neither high entropy nor a second visual observation alone can modify the answer. A repair is committed only after a revisable fact is localized, answer-blind evidence directly contradicts it, and the locally replayed output passes the final safeguards. Counterpoise therefore exchanges a small number of targeted verification calls for avoiding repeated full-trajectory sampling and unnecessary global regeneration, providing a controllable trade-off between reasoning reliability and additional inference cost.

V. EXPERIMENTS

A. Experimental Setup

We conducted an extensive evaluation of 19 representative commercial and open-weight LMMs on Domino-Bench. All models were evaluated using their official implementations or publicly available APIs. For systematic comparison, we organized the evaluated models into four categories: (1) commercial API models, (2) open-weight models with at least 70B parameters, (3) open-weight models with 30B–69B parameters, and (4) open-weight models with 7B–29B parameters. This grouping allows us to compare commercial and open-weight systems while examining performance differences across parameter scales and model generations.

Domino-Bench contains open-ended multimodal reasoning problems from six theory-intensive disciplines: mathematics, physics, engineering, chemistry, geography, and biology. Unless otherwise specified, we followed the official inference configurations of each model and used deterministic decoding with the temperature set to zero. Each model was required to generate a complete reasoning process and final answer without access to predefined answer candidates.

For process-level evaluation, we provided the evaluator with the question, the annotated key intermediate claims, and the reasoning process generated by the evaluated LMM. The evaluator then determined whether the generated reasoning supported each claim. For perception claims, the original image was additionally provided as visual evidence. We employed GPT-5.4-mini as the evaluator because of its strong logical reasoning and visual understanding capabilities. Based on the resulting claim labels, claim categories, and reasoning dependencies, we computed knowledge, perception, and reasoning claim accuracy, overall claim accuracy, and DAG-HS. We used identical evaluation prompts, answer-parsing rules, and process-level evaluation protocols across all models. Empty generations, unparseable final answers, and missing evaluator outputs were excluded only from the corresponding metric calculations.

B. Main Results

1) *Overview of Model Performance:* Table VI presents the main results of 19 representative LMMs on Domino-Bench. Overall, Domino-Bench poses a substantial challenge to current models. The two best-performing models, GPT-5.6-sol-xhigh and Gemini-3.1-Pro, achieve average answer accuracies of only 67.18% and 65.96%, respectively. Moreover, the evaluated models exhibit a broad distribution of overall scores, indicating that performance on Domino-Bench remains far from saturated. These results demonstrate that the benchmark can effectively differentiate the open-ended multimodal scientific reasoning capabilities of current LMMs while leaving considerable room for future improvement.

A noticeable performance gap also exists between commercial API models and open-weight models. Commercial models generally occupy the top positions in Table VI, whereas Gemma-4-31B-it, the best-performing open-weight model, achieves an average accuracy of 32.89%, substantially below GPT-5.6-sol-xhigh and Gemini-3.1-Pro. Notably, even GLM-5V-Turbo, which ranks relatively lower among the evaluated commercial models, achieves an average accuracy of 42.13%, outperforming all evaluated open-weight models. These results provide initial evidence that commercial models retain an overall advantage in complex multimodal understanding and open-ended multi-step reasoning, while a considerable capability gap remains between open-weight models and frontier commercial systems. 2) *Performance Across Disciplines:* As shown in Table VI, even the best-performing model, GPT-5.6-sol-xhigh, exhibits substantial performance variations across disciplines. It achieves accuracies of 86.76%, 74.36%, and 72.82% in mathematics, physics, and engineering, respectively, whereas its performance decreases to 48.74% in chemistry and 47.70% in geography. Since different disciplines emphasize distinct capabilities, including logical deduction, visual perception, domain-specific knowledge, and spatial understanding, these cross-disciplinary variations indicate that current models have yet to achieve balanced scientific reasoning capabilities across domains.

TABLE III
CROSS-BENCHMARK COMPARISON BETWEEN DOMINO-BENCH AND MATHVISTA. MODELS ARE ORDERED BY THEIR AVERAGE ACCURACY ON DOMINO-BENCH.

Model	Domino-Bench	MathVista
Qwen3-VL-32B	27.68	83.8
Metis-RISE-72B	26.15	80.4
GLM-4.1V-9B	23.70	80.7
VL-Rethinker-72B	23.42	80.4
InternVL3-78B	23.34	79.0
InternVL3-38B	22.30	75.1
Qwen2.5-VL-72B	21.81	74.2
Qwen3-VL-8B-Instruct	21.74	77.2
InternVL3-14B	19.53	75.1
MiniCPM-V-4.5	18.58	79.9
NVLM-D-72B	15.27	65.2
LLaVA-CoT-11B	12.14	54.8
InternVL3-2B	11.99	57.0
LLaVA-v1.6-34B	7.97	46.5

C. Error Taxonomy and Hallucination Propagation

D. Validation of the Evaluation Framework and DAG-HS

E. Reliability and Human Agreement

To examine the reliability of our process-level evaluation, we evaluate both the quality of the CoT-to-DAG transformation and the agreement between automatic and human judgments of claim correctness.

DAG construction quality. We select two models from each of the four previously defined model groups, resulting in eight representative models with diverse capability levels. We then perform joint stratified sampling over the combinations of six disciplines and four difficulty levels in Domino-Bench. Questions are randomly sampled within each discipline–difficulty stratum to construct a shared evaluation set of 200 questions. All eight models are evaluated on the same question set, yielding 1,600 reasoning trajectories and their corresponding DAGs.

We employ GPT-5.4-mini to jointly inspect each original CoT and its corresponding DAG and assess the transformation quality along six dimensions on a scale from 0 to 10. The first three dimensions focus on individual nodes. *Information fidelity* measures whether each node accurately preserves the meaning of its corresponding content in the original CoT without distortion or unsupported additions. *Claim atomicity* examines whether each node expresses a single, self-contained, and verifiable claim. *Node-type correctness* evaluates whether each node is correctly categorized as a knowledge, perception, or reasoning claim.

The next two dimensions evaluate the logical relations between nodes. *Dependency completeness* measures whether the necessary premise–conclusion relations in the original CoT are captured by the DAG. *Dependency correctness* assesses whether the extracted edges and their directions accurately represent the reasoning dependencies between nodes. Finally,

TABLE IV
MODEL-WISE RELIABILITY EVALUATION OF THE CoT-TO-DAG TRANSFORMATION.

Model	Information Fidelity		Claim Atomicity		Node-Type Correctness		Dependency Completeness		Dependency Correctness		Information Coverage	
	Avg.	Pass (%)	Avg.	Pass (%)	Avg.	Pass (%)	Avg.	Pass (%)	Avg.	Pass (%)	Avg.	Pass (%)
GPT-5.6-sol-xhigh	–	–	–	–	–	–	–	–	–	–	–	–
GLM-5V-Turbo	–	–	–	–	–	–	–	–	–	–	–	–
Metis-RISE-72B	–	–	–	–	–	–	–	–	–	–	–	–
NVLM-D-72B	–	–	–	–	–	–	–	–	–	–	–	–
Gemma-4-31B-it	–	–	–	–	–	–	–	–	–	–	–	–
InternVL3.5-38B	–	–	–	–	–	–	–	–	–	–	–	–
GLM-4.1V-9B	–	–	–	–	–	–	–	–	–	–	–	–
LLaVA-CoT-11B	–	–	–	–	–	–	–	–	–	–	–	–

Note. Each model is evaluated on the same set of 200 questions. Scores range from 0 to 10. **Avg.** denotes the average score, while **Pass** denotes the percentage of DAGs receiving a score of at least 6.

information coverage compares the complete node set against the original CoT and measures the extent to which the nodes collectively preserve its substantive reasoning information.

For each dimension, we report the average score over all 1,600 DAGs and the corresponding pass rate. A DAG is considered to pass a dimension if it receives a score of at least 6. As summarized in Table IV, these dimensions collectively evaluate node extraction, dependency reconstruction, and the overall coverage of the original reasoning content.

Human agreement on claim judgments. We further examine the reliability of the claim-correctness labels assigned by GPT-5.4-mini. From the shared set of 200 questions, we randomly sample the same subset of 50 questions for all eight models. We manually review the corresponding model outputs, resulting in 400 reasoning trajectories and all claims contained within them.

Two human annotators independently determine the correctness of each claim without access to the labels produced by the automatic evaluator. Their judgments are based on the question, the original reasoning context, and the relevant supporting evidence. For perception claims, the original image is additionally provided. Disagreements between the two annotators are resolved by a third annotator.

We first report the raw agreement and Cohen’s κ between the two human annotators to evaluate the stability of human judgments. We then use the adjudicated human labels as references and compute the agreement between GPT-5.4-mini and the human judgments. Because different reasoning trajectories may contain different numbers of claims, all agreement statistics are calculated at the claim level rather than the trajectory level. The results are reported in Table V.

Together, these experiments evaluate the reliability of both the CoT-to-DAG transformation and the automatic claim-correctness judgments used in our process-level evaluation framework.

F. Evaluation of Counterpoise

VI. TEXT

For some of the remainder of this sample we will use dummy text to fill out paragraphs rather than use live text that may violate a copyright.

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TABLE V
MODEL-WISE HUMAN VERIFICATION OF CLAIM-CORRECTNESS JUDGMENTS.

Model	Claim Correct Rate		Judgment Agreement		
	Human	Judge	H–H	Cohen’s κ	J–H
GPT-5.6-sol-xhigh	–	–	–	–	–
GLM-5V-Turbo	–	–	–	–	–
Metis-RISE-72B	–	–	–	–	–
NVLM-D-72B	–	–	–	–	–
Gemma-4-31B-it	–	–	–	–	–
InternVL3.5-38B	–	–	–	–	–
GLM-4.1V-9B	–	–	–	–	–
LLaVA-CoT-11B	–	–	–	–	–

Note. Each model is evaluated on the same subset of 50 questions. All statistics are computed at the claim level. **Human** and **Judge** denote the proportions of claims judged as correct according to the adjudicated human annotations and GPT-5.4-mini, respectively. **H–H** denotes the raw agreement between the two independent human annotators. Cohen’s κ measures inter-annotator agreement after accounting for chance agreement. **J–H** denotes the agreement between GPT-5.4-mini and the adjudicated human labels. All results except Cohen’s κ are reported as percentages.

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TABLE VI
MAIN RESULTS ON DOMINO-BENCH.

Model	Type	Answer Accuracy by Discipline						Avg. Acc.	Claim Accuracy				DAG-HS
		Math	Physics	Engineering	Chemistry	Geography	Biology		Knowledge	Perception	Reasoning	Overall	
Commercial API models													
GPT-5.6-sol-xhigh	Native	86.76	74.36	72.82	48.74	47.70	65.56	67.18	—	—	—	—	—
Gemini-3.1-Pro	Native	<u>86.30</u>	75.24	75.80	29.23	57.47	<u>64.24</u>	<u>65.96</u>	—	—	—	—	—
Grok-4.5-high	Native	75.55	63.34	63.48	48.74	32.42	54.30	57.29	—	—	—	—	—
Qwen3.7-plus	Native	77.33	67.20	64.54	36.50	30.56	54.75	55.84	—	—	—	—	—
Kimi-2.7-code	Native	71.87	55.13	66.45	<u>36.89</u>	37.02	59.16	55.32	—	—	—	—	—
MiniMax-M3	Native	54.41	40.58	45.65	30.52	34.29	61.81	44.92	—	—	—	—	—
GLM-5V-Turbo	Native	54.34	42.53	46.07	26.89	23.63	60.26	42.13	—	—	—	—	—
Open-weight models with $\geq 70B$ parameters													
Metis-RISE-72B	Native	35.63	<u>18.77</u>	26.75	12.48	15.05	<u>45.03</u>	26.15	—	—	—	—	—
VL-Rethinker-72B	Native	<u>32.02</u>	19.09	<u>23.14</u>	9.06	13.68	42.16	<u>23.42</u>	—	—	—	—	—
InternVL3-78B	Native	29.97	17.15	22.29	9.21	<u>14.70</u>	47.02	23.34	—	—	—	—	—
Qwen2.5-VL-72B	Native	27.53	18.18	<u>23.14</u>	<u>9.84</u>	11.80	43.05	21.81	—	—	—	—	—
NVLM-D-72B	Native	14.29	9.06	12.53	9.21	9.81	43.71	15.27	—	—	—	—	—
Open-weight models with 30B–69B parameters													
Gemma-4-31B-it	Native	45.98	31.39	38.22	20.95	<u>13.81</u>	44.81	32.89	—	—	—	—	—
Qwen3-VL-32B	Native	<u>33.91</u>	<u>23.62</u>	<u>33.33</u>	<u>18.28</u>	16.04	<u>42.48</u>	<u>27.68</u>	—	—	—	—	—
InternVL3.5-38B	Native	29.47	18.18	23.78	9.66	<u>13.81</u>	40.84	22.61	—	—	—	—	—
LLaVA-v1.6-34B	Concat.	5.50	4.21	5.31	4.90	3.98	31.57	7.97	—	—	—	—	—
Open-weight models with $< 30B$ parameters													
GLM-4.1V-9B	Native	36.54	17.15	21.23	<u>10.55</u>	7.59	44.37	23.70	—	—	—	—	—
Qwen3-VL-8B-Instruct	Native	<u>28.57</u>	<u>20.06</u>	<u>20.38</u>	10.25	10.93	42.16	<u>21.74</u>	—	—	—	—	—
InternVL3.5-14B	Native	22.99	14.56	18.68	11.29	13.68	42.60	20.16	—	—	—	—	—
InternVL3.5-8B	Native	25.86	15.53	15.92	9.81	11.57	42.83	20.14	—	—	—	—	—
InternVL3-14B	Native	24.38	14.61	16.35	6.98	<u>12.19</u>	<u>44.81</u>	19.53	—	—	—	—	—
MiniCPM-V-4.5	Native	23.32	11.00	16.99	8.92	8.45	45.03	18.58	—	—	—	—	—
Gemma-4-12B-it	Native	27.01	22.01	12.10	5.79	7.59	27.59	17.29	—	—	—	—	—
LLaVA-CoT-11B	Concat.	10.10	7.44	8.07	7.13	7.84	40.18	12.14	—	—	—	—	—
InternVL3-2B	Native	11.25	8.74	8.49	5.50	8.20	36.20	11.99	—	—	—	—	—
NEO1.0-9B	Native	7.22	6.47	5.10	3.27	4.35	30.46	8.32	—	—	—	—	—

Note. All results are reported as percentages. Answer accuracy is evaluated on the selected medium, medium-high, and high-difficulty subsets, comprising approximately 3.9K samples. The exact number of evaluated samples varies slightly across models because of generation failures, empty responses, or missing judge outputs. **Type** indicates whether a model follows a concatenation-based multimodal architecture or a native multimodal architecture. **Math**, **Physics**, **Engineering**, **Chemistry**, **Geography**, and **Biology** denote the six disciplines in Domino-Bench. **Avg. Acc.** denotes the micro-averaged final-answer accuracy over all evaluated samples. **Knowledge**, **Perception**, and **Reasoning** denote claim-level accuracy for knowledge, perception, and reasoning claims, respectively. **DAG-HS** denotes the proposed dependency-aware hierarchical support score computed over the reasoning DAG. The best and second-best results within each model category are highlighted in **bold** and underlined, respectively. Tied best or second-best results are highlighted equally. Models within each category are ordered by their average final-answer accuracy.

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VII. TABLES

Note that, for IEEE-style tables, the `\caption` command should come BEFORE the table. Table captions use title case. Articles (a, an, the), coordinating conjunctions (and, but, for, or, nor), and most short prepositions are lowercase unless they are the first or last word. Table text will default to `\footnotesize` as the IEEE normally uses this smaller font for tables. The `\label` must come after `\caption` as always.

VIII. CONCLUSION

The conclusion goes here.

TABLE VII
AN EXAMPLE OF A TABLE

One	Two
Three	Four

ACKNOWLEDGMENTS

This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

APPENDIX PROOF OF THE ZONKLAR EQUATIONS

Use `\appendix` if you have a single appendix: Do not use `\section` anymore after `\appendix`, only `\section*`. If you have multiple appendixes use `\appendices` then use `\section` to start each appendix. You must declare a `\section` before using any `\subsection` or using `\label` (`\appendices` by itself starts a section numbered zero.)

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